UBSTrack: Unified Band Selection and Multimodel Ensemble for Hyperspectral Object Tracking

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***Abstract*— Hyperspectral object tracking (HOT) is notably challenging due to the high-dimensional nature of the data and the necessity of seamlessly integrating spectral, spatial, and temporal information. Traditional methods often emphasize detection-based or tracking-based networks, each leveraging their inherent strengths but overlooking the potential advantages of a combined approach, leading to suboptimal performance in complex, real-world scenarios. Furthermore, this challenge is amplified by the variability of spectral bands across datasets, making the maintenance of consistent tracking performance complicated. To address these issues, we propose a novel, unified approach that merges adaptive band selection with a multimodel ensemble strategy. We introduce a local and global attention-based unified band selection (UBS) technique that identifies the most informative three bands from any dataset, significantly reducing data complexity while preserving critical spectral and spatial information. This UBS method employs spectral independence, allowing it to process hyperspectral video frames with any number of bands as input, ultimately generating a three-band pseudocolor image. This is coupled with a multimodel ensemble framework, utilizing a local and global attention-based appearance module. The module selects the optimal candidate by computing the similarity between the proposals generated by the base models and historical frames. Experimental results show that our approach, UBSTrack, achieves state-of-the-art performance, delivering robust and accurate tracking under different real-world challenging scenarios. The code of UBSTrack is available at the following link: https://github.com/aamin0102/UBSTrack/**

***Index Terms*— Band selection, ensemble learning, fusion, hyperspectral, object tracking, transformer, YOLO.**

I. Introduction

HYPERSPECTRAL imaging (HSI) has emerged as a powerful tool in diverse real-world applications, including precision agriculture [1], environmental monitoring [2], disaster management [3], and urban planning [4]. In particular, unmanned aerial vehicles (UAVs) equipped with lightweight hyperspectral sensors are increasingly used for tasks such as crop health monitoring [5], forest fire detection [6], and flood damage assessment [7], offering ultrahigh-resolution and flexible data collection. Similarly, airborne platforms enable regional-scale surveys for geological mapping, vegetation analysis, and urban infrastructure inspection [8], while spaceborne hyperspectral sensors provide global coverage for monitoring large-scale phenomena like deforestation, land-use change, and climate impacts [9]. These platforms produce vast amounts of hyperspectral data, which, if processed effectively, can deliver highly accurate and actionable insights for decision-makers.

One promising area where HSI offers a transformative impact is object tracking. Traditional object tracking systems, often reliant on RGB imagery, face significant challenges in complex environments where objects exhibit similar appearances, dynamic backgrounds, or varying illumination [10]. HSIs address these limitations by capturing detailed spectral information across hundreds of contiguous bands, enabling the precise differentiation of materials and objects based on their unique spectral signatures [11]. This capability makes hyperspectral object tracking (HOT) particularly advantageous for UAV-based applications, such as monitoring crop diseases, tracking fire boundaries, or assessing flood extents by distinguishing water from vegetation and debris [12]. Furthermore, the spectral richness of HSI allows for robust tracking even in complex environments with background clutter and occlusions [10].

Despite its potential, effective HOT remains a challenging task due to the high-dimensional nature of hyperspectral data, redundancy across spectral bands, and the scarcity of large annotated datasets for training deep learning models. Many HOT methods have been proposed to address these challenges, utilizing both conventional techniques and deep learning (DL) approaches. Conventional methods often rely on hand-crafted features such as SSHMG [10], HOMG [13], and 3-D HOG [14], which are typically combined with RGB-based trackers to enhance performance. Some approaches also integrate hand-crafted features with deep learning-based representations to improve object representation [15]. In contrast, DL-based methods make significant advancements by automatically learning complex spatial–spectral representations. These methods employ a range of strategies, including detection-based models leveraging classifiers [16], [17], Siamese networks [18], [19], and transformer-based architectures [20]. Additionally, some methods integrate RGB-HSI fusion-based [21] or

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RGB-HSI prompt-based models [22]. The adaptability of DL models to varying tracking scenarios makes them effective in capturing intricate patterns and subtle differences between targets and backgrounds.

However, DL approaches are data-hungry and require substantial annotated hyperspectral datasets to train effectively. The scarcity of such datasets poses a significant limitation, hindering the full potential of these approaches. To address this challenge, several band selection methods have been proposed, leveraging either hand-crafted features or DL techniques to convert hyperspectral images into three-band pseudocolor images. This conversion allows deep networks, pre-trained on large RGB video datasets, to be effectively applied to hyperspectral video data. Conventional methods like Gaussian Information Entropy [23] and Principal Component Analysis [24] have been employed to reduce data volume but often fail to fully exploit hyperspectral data. Therefore, attention mechanisms [19], hierarchical strategies [11], convolutional networks [25], and end-to-end encoder–decoder architectures [26] have been introduced.

Despite recent advancements, existing band selection techniques for HOT suffer from key limitations that hinder their generalization and practical deployment. Many approaches rely on fixed numbers of spectral bands [27] or are tailored to specific spectral characteristics [11], [17], [25], [26], [28], making them highly dataset-dependent. As a result, models often need to be retrained or customized for each dataset, which varies in both the number and distribution of spectral bands. This lack of adaptability leads to overfitting and poor performance when applied to unseen or diverse data. Furthermore, the limited availability of large, diverse hyperspectral datasets exacerbates this issue, preventing these methods from learning robust, transferable representations. Consequently, their performance degrades significantly in real-world scenarios that demand consistent tracking across varying conditions and sensor configurations.

Moreover, detection-based and tracking-based methodologies serve as the foundation of most HOT frameworks, yet each exhibits distinct strengths and limitations. Detection-based methods are effective at handling occlusions and abrupt appearance changes by re-identifying objects in each frame, providing strong per-frame localization [29]. However, this frame-wise processing lacks temporal continuity, often resulting in fragmented or inconsistent tracking. In contrast, tracking-based methods leverage temporal coherence by predicting object positions based on historical data, offering smoother and more stable tracking for objects with consistent appearances [30]. Still, they struggle with significant appearance variations, long-term occlusions, or object reentry, leading to drift or failure [31]. These complementary strengths and weaknesses highlight the need for a hybrid approach. A robust HOT framework should not only unify band selection but also incorporate an innovative ensemble strategy that effectively integrates both detection- and tracking-based models. Such integration can balance temporal stability with adaptability, enhancing overall accuracy and robustness across dynamic and challenging scenarios.

Therefore, in this article, we propose UBSTrack, a novel and comprehensive framework designed to address the core challenges of HOT. First, we design a dataset-independent adaptive band selection method that leverages local and global attention mechanisms to dynamically identify the most informative spectral bands. Unlike previous methods that rely on fixed-band strategies or dataset-specific tuning, our approach performs video-level band selection, applying a consistent set of selected bands across all frames, thereby enhancing generalization and robustness. This adaptability allows the technique to be applied across any datasets with differing numbers of spectral bands and spectral characteristics. Second, we propose a multimodel ensemble strategy that combines detection-based and tracking-based networks in a novel way. Our ensemble framework incorporates an appearance learning module, which computes the similarity between candidate proposals and the target's historical appearance to select the optimal tracking output. This mechanism ensures temporal consistency while retaining spectral and spatial fidelity. Furthermore, a GATE mechanism is incorporated to control the flow of features to align with the target's appearance in previous frames, ensuring a coherent representation over time. This selective gating enhances UBSTrack's adaptability and efficiency, resulting in improved tracking accuracy and robustness in complex real-world scenarios. The contributions of the article are listed below.

1) We propose UBSTrack, a unified framework that introduces adaptive, dataset-agnostic band selection and a novel ensemble approach that integrates detection and tracking models using appearance-based matching and gated temporal consistency.
2) Our dynamic band selection method effectively identifies the most informative spectral bands independently of specific datasets, ensuring adaptability across various hyperspectral data scenarios.
3) We develop a local-global appearance learning module and a GATE mechanism to ensure temporal coherence and selective feature refinement during tracking.
4) We conduct extensive experiments on the HOT2024 dataset, showing that UBSTrack outperforms existing approaches, especially in challenging real-world scenarios.

The structure of the article is as follows. Section II provides a detailed review of the related literature. In Section III, we introduce our proposed tracking framework, detailing its components and methodology. Section IV covers the experimental results, discussing the datasets used, evaluation metrics applied, and the performance of UBSTrack across different tracking scenarios. Lastly, conclusions are presented in Section V.

II. Literature Review

In this section, we focus on the key strategies and methodologies that have been developed for HOT. We structure our review into three main parts: band selection strategies, conventional HOT methods, and deep learning-based HOT approaches.

A. Band Selection Strategies

Band selection is a vital preprocessing step in HOT, designed to reduce the high dimensionality of data while retaining the most significant spectral information. One of the

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primary challenges in HOT is the limited availability of diverse and comprehensive hyperspectral datasets, often necessitating the use of RGB datasets as substitutes. However, RGB data fails to capture the unique and detailed spectral characteristics inherent in HSI. Additionally, the high correlation among spectral bands adds further complexity, making it difficult to select bands without introducing redundancy or omitting essential spectral features.

To address these issues, various conventional band selection methods have been employed. Some trackers utilize genetic algorithms [32], [33], while others implement techniques such as the Optimum Index Factor [20] or Gaussian Information Entropy [23]. Principal Component Analysis [24] has also been widely used to reduce data volume, and certain methods [16], [18] have leveraged cross-band dissimilarity information to enhance selection. Despite their utility, these traditional approaches often fall short of fully exploiting the rich information provided by hyperspectral data.

Therefore, DL-based band selection techniques have been introduced to address the limitations of conventional methods. Some approaches employ attention mechanisms [17], [19] to sort and select the most prominent bands, while other methods utilize a hierarchical attention strategy [11] for more refined band selection, allowing for a structured approach to handling spectral data. Additionally, a few techniques [25], [34] have leveraged convolutional networks to map the HSI cube into the RGB domain, simplifying data representation while retaining key spectral features. End-to-end encoder–decoder architectures [26], [35] have also been utilized for band selection, providing automated feature extraction and dimensionality reduction within a unified framework.

However, these deep learning-based methods are often spectral-specific, limiting their generalization ability across diverse hyperspectral datasets, which may vary in spectral bands, resolutions, or ranges. Therefore, there is a pressing need for a unified band selection (UBS) technique capable of overcoming this data dependency problem, enabling consistent performance across datasets with varying spectral properties and ensuring broader applicability in HOT.

B. Conventional HOT

Conventional HOT methods used correlation filters (CFs) to establish relationships between the target object's features and its surrounding context within hyperspectral videos. One of the pioneering works in this domain is by Qian et al. [36], who integrated convolutional networks with a kernelized CF (KCF) [37] framework for HOT. Their method employed normalized 3-D feature cubes as fixed convolution filters to encode local spectral–spatial features, which KCF then used for tracking. Building on this, Xiong et al. [10] introduced an approach that extracted material information from the target object using spectral–spatial histograms of multidimensional gradients (SSHMGs) to represent 3-D local spectral–spatial structures and fractional abundances of materials.

Subsequent CF-based methods have sought to improve the extraction of discriminative features. For example, Chen et al. [38] utilized spatial–spectral convolution kernels in the Fourier transform domain to extract spatial–spectral features. In contrast, Chen et al. [13] proposed a histogram of oriented mosaic gradient (HOMG) descriptors for better feature representation. Furthermore, Tang et al. [15] introduced adaptive spatial–spectral discriminant analysis for enhanced feature extraction. Hou et al. [39] developed a CF-based framework employing a tensor sparse CF with a spatial–spectral weighted regularizer, reducing spectral differences in inhomogeneous backgrounds and penalizing filter templates based on spectral dissimilarity.

These CF-based methods have laid a solid foundation for hyperspectral tracking by leveraging spectral–spatial information, but they often face challenges in adapting to complex scenarios involving large appearance variations or occlusions.

C. Deep-Learning-Based HOT

DL-based HOT methods have emerged as powerful alternatives to traditional CF techniques, offering the ability to automatically learn complex spatial and spectral feature representations. These methods leverage the power of convolutional neural networks (CNNs) and other advanced architectures, such as Siamese networks and transformers, to capture the intricate patterns present in hyperspectral data, making them particularly effective in challenging real-world scenarios.

Several methods have successfully adapted pre-trained RGB trackers to the hyperspectral domain. Notably, methods such as TLHT [25], BABS [16], BAE-Net [17], and SST-Net [40] employed classifier-based architectures. TLHT [25] utilized a pre-trained VGG-M network to transfer learned features from RGB data to hyperspectral data, while MDNET [41] facilitated online tracking, though it faced issues with appearance variations and class imbalance. To address these limitations, methods like BABS [16], BAE-Net [17], and SST-Net [40] incorporated VITAL [42], which leverages adversarial learning and a cost-sensitive loss function to improve robustness. Recently, Hy-Tracker by Islam et al. [11] combined YOLO for real-time detection with a classifier and GRU, enabling better handling of background clutter and maintaining temporal consistency in HOT.

Siamese networks have also been adapted for HOT, leveraging the rich spectral information in hyperspectral data. For instance, SiamMRAAN [43] used a five-branch SiamFC [44] network with a novel loss function to boost performance in HOT. D-Siam [45] introduced a dual-branch SiamFC architecture, combining response maps from multiple frames through an adaptive fusion strategy to enhance tracking accuracy. Furthermore, methods such as AD-SiamRPN [32] and BS-SiamRPN [33] applied transfer learning and dynamic template updates to handle appearance changes in hyperspectral data, making SiamRPN [46] more effective. Other approaches, including BAHT [35], GIE [23], and SiamHT [47], extended pre-trained RGB-based SiamBan [48] by incorporating spectral-semantic features to better exploit hyperspectral information.

Recent advances have also seen the application of transformer-based architectures in HOT. RawTrack [49] and HELIOS [50] adapted RGB-based OSTrack [51] for HOT, while Trans-HST [52] and AHS-Net [53] utilized multiscale attention and vision transformers (ViTs [54]) to capture long-range dependencies and complex feature interactions. DETR [55]-based trackers such as HSP-Track [20] and SPIRIT [56] refined object localization, while

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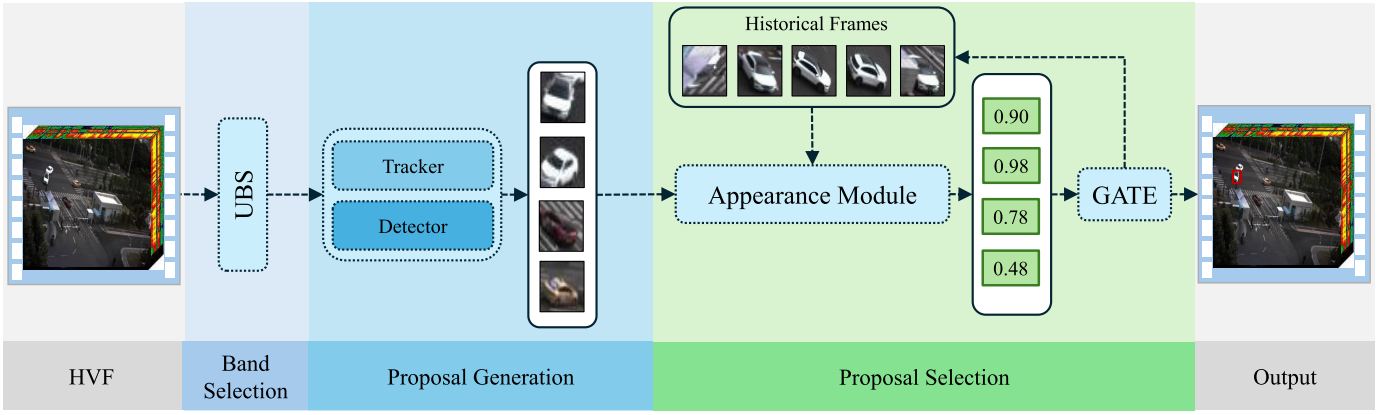


Fig. 1. Overview of the UBSTrack model framework. The framework consists of three main stages: 1) band selection, where hyperspectral video frames are processed using a dataset-independent UBS method to identify the three most informative bands and create a pseudocolor image; 2) proposal generation, which employs a tracker and a detector to generate candidate proposals from the pseudocolor image; and 3) proposal selection, where the appearance module calculates similarity scores between model proposals and historical frames of the target to ensure temporal consistency and select the most reliable proposal. The selected candidate utilizes the GATE mechanism to regulate feature flow and update historical frame representations, enhancing tracking accuracy and robustness.

SENSE [57] used cross-modality fusion for enhanced performance. Trans-DAT [58] further improved tracking robustness with transformer-based feature fusion. More recently, prompt-based fusion models, such as VP-HOT [22], SPTrack [59], SP-HST [34], and SIPHOT [60], have been introduced, integrating RGB and hyperspectral data by embedding both into token representations. In these approaches, prompts derived from hyperspectral data are injected into the RGB data stream, enriching the feature representations at multiple layers, which leads to enhanced spatial–spectral integration and improved tracking accuracy.

Despite these advancements, current DL-based HOT methods often face limitations in fully integrating spatial, spectral, and temporal features. Detection models may overlook critical spectral details, prompt-based models can struggle with occlusions, and tracking networks often lose accuracy in cluttered environments. These challenges limit existing HOT methods' overall robustness and effectiveness in complex real-world scenarios. Therefore, there is a pressing need for a more comprehensive framework that can effectively integrate the complementary strengths of multiple specialized models to address these limitations and improve tracking performance across a wide range of conditions.

III. UBSTrack Framework

The UBSTrack framework is designed to address the dataset-dependency issues inherent in existing band selection techniques and to leverage the complementary strengths of detection and tracking models. By employing a local and global attention-based appearance module, UBSTrack ensures robust HOT. The schematic of the framework is shown in Fig. 1, illustrating its three main modules: UBS, proposal generation, and proposal selection. A detailed description of each module in the UBSTrack framework is provided in this section.

A. Unified Band Selection

The UBS module is designed to overcome the dataset-dependency issues prevalent in existing band selection methods. Unlike existing techniques that are often designed for specific spectral datasets and struggle to generalize across different spectral properties, the UBS method is adaptable and independent of individual datasets. It leverages local and global attention mechanisms to dynamically identify the three most informative spectral bands, reducing data dimensionality while preserving essential spectral information.

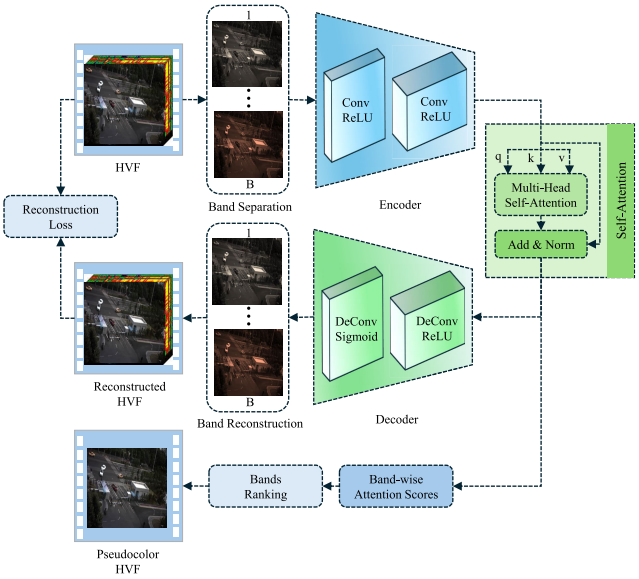


Fig. 2. Architecture of UBS employing local and global attention mechanisms.

The architecture of the proposed UBS model, illustrated in Fig. 2, integrates local and global attention mechanisms to facilitate dataset-independent band selection for HOT. The process begins with the input of a hyperspectral video frame (HVF) cube $\mathbf{X} \in \mathbb{R}^{H\times W\times B}$, where H, W, and B represent the height, width and spectral bands, respectively. The HVF is passed through a band separation module, dividing the spectral cube into individual band slices $\mathbf{X}_b \in \mathbb{R}^{H\times W\times 1}$ for $b = 1, \ldots, B$. Each band $\mathbf{X}_b$ is processed through a local attention-based convolutional encoder comprising two

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Conv + ReLU components, represented as

$$\mathbf{F}_b = \text{Encoder}(\mathbf{X}_b) \tag{1}$$

where Encoder(.) denotes the local-attention-based convolutional operations, producing feature maps $\mathbf{F}_b$. Each feature map $\mathbf{F}_b$ is subsequently passed to a multihead self-attention mechanism to capture global contextual relationships and also attention weights across all spectral bands. The self-attention mechanism operates on these encoded features and is defined as

$$\hat{\mathbf{F}}_b = \text{Self-Attention}(\mathbf{F}_b) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{2}$$

where $\mathbf{Q} = \mathbf{W}_Q\mathbf{F}_b$, $\mathbf{K} = \mathbf{W}_K\mathbf{F}_b$, and $\mathbf{V} = \mathbf{W}_V\mathbf{F}_b$. Here, $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ are learnable weight matrices, and d_k is the dimensionality of the key vectors.

The global contextual attention feature of each band $\hat{\mathbf{F}}_b$ is then fed into a decoder comprising DeConv + ReLU and DeConv + Sigmoid layers to reconstruct the individual band $\hat{\mathbf{X}}_b$

$$\hat{\mathbf{X}}_b = \text{Decoder}\left(\hat{\mathbf{F}}_b\right) \tag{3}$$

where Decoder(.) represents the decoding process applied to each band attention feature map. After that, all the reconstructed bands are reformed to reproduce the reconstructed HVF frame, $\hat{\mathbf{X}}$.

Simultaneously, the attention weights generated for each band are used to rank the importance of the bands

$$\alpha_b = \frac{1}{H' \times W'} \sum_{i=1}^{H'} \sum_{j=1}^{W'} \text{Self-Attention}(\mathbf{F}_b)_{i,j} \tag{4}$$

where i and j index the spatial dimensions of the feature map, and H' and W' are the height and width of the attention feature map $\hat{\mathbf{F}}_b$. The bands are then ranked based on these attention scores, and the top three bands are selected to create a pseudocolor image.

B. Proposal Generation

Proposal generation aims to produce a diverse set of candidate proposals by leveraging two complementary models: the tracker OSTrack [51] and the detector YOLO [61]. Each model specializes in different feature representations to ensure a comprehensive understanding of the target object. Given a pseudocolor video frame $\mathbf{X}_{\text{PC}_t} \in \mathbb{R}^{H\times W\times 3}$, where H and W represent height and width of spectral bands, respectively, extracted from the original HVF $\mathbf{X}_t \in \mathbb{R}^{H\times W\times B}$ using the UBS module, these two models process the data to generate candidate proposals.

The tracker is designed for robust tracking and utilizes spatial–temporal features from $\mathbf{X}_{\text{PC}_t}$ to maintain appearance consistency and track the object's movement across frames. This process can be formulated as

$$\mathcal{P}_{\text{T}} = f_{\text{T}}\left(\mathbf{X}_{\text{PC}_t}\right) \tag{5}$$

where $\mathcal{P}_{\text{T}}$ denotes the proposal generated by the Tracker. Simultaneously, the detection-based model processes the pseudocolor frame $\mathbf{X}_{\text{PC}_t}$ to identify potential target regions and generates bounding box proposals based on pre-trained detection features

$$\mathcal{P}_{\text{D}} = f_{\text{D}}\left(\mathbf{X}_{\text{PC}_t}\right) \tag{6}$$

where $\mathcal{P}_{\text{D}}$ denotes the set of N_1 proposals generated by the detectors. The resulting candidate proposals from both models are combined into a unified proposal set

$$\mathcal{P} = \mathcal{P}_{\text{T}} \bigcup \mathcal{P}_{\text{D}}. \tag{7}$$

Here, $\mathcal{P}$ is the combined candidate proposals from both tracker and detector, where the total number of proposals is $N = 1 + N_1$.

C. Proposal Selection

The goal of proposal selection is to identify the most reliable proposal by measuring the similarity between candidate proposals and the target's appearance in historical frames. Each candidate proposal, $\mathbf{p}_i \in \mathcal{P}$, is compared with all historical frames using the appearance module, resulting in a set of similarity scores. The similarity scores are then averaged for each proposal, and the proposal with the highest average similarity score is selected.

Appearance module: The appearance module in UBSTrack is designed to compute the similarity between the candidate proposals and the historical appearance of the target, thereby aiding in the selection of the most reliable proposal. The module processes both the candidate proposal and the historical target frames using the same feature extraction and attention mechanisms to ensure consistency. The schematic of the appearance module is shown in Fig. 3. This design is inspired by [62], where shared feature representations and attention mechanisms were shown to be effective in modeling appearance similarity over time.

Given a candidate proposal image $\mathbf{p}_i$ and a historical frame image $\mathbf{h}_j$, where $i = 1, 2, \ldots, N$ (number of proposals) and $j = 1, 2, \ldots, M$ (number of historical frames), the feature extraction process can be defined as

$$\mathbf{F}_{p_i} = \text{ResNet18}(\mathbf{p}_i), \mathbf{F}_{h_j} = \text{ResNet18}\left(\mathbf{h}_j\right) \tag{8}$$

where $\mathbf{F}_{p_i} \in \mathbb{R}^{H\text{x}W\text{x}3}$ and $\mathbf{F}_{h_j} \in \mathbb{R}^{H\text{x}W\text{x}3}$ denote the extracted feature maps for the proposal and historical frame, respectively.

The extracted feature maps are then divided into four smaller patches $\mathbf{F}_{\text{PA}} = \{\mathbf{F}_{\text{PA}_1}, \mathbf{F}_{\text{PA}_2}, \mathbf{F}_{\text{PA}_3}, \mathbf{F}_{\text{PA}_4}\}$. Each patch is passed through an attention mechanism composed of two sequential self-attention layers to capture global contextual relationships. The first self-attention operation is applied using (2) to each patch is defined as

$$\mathbf{F}^{(1)}_{\text{PA}_i} = \text{Self-Attention}\left(\mathbf{F}_{\text{PA}_i}\right) \tag{9}$$

where $\mathbf{F}^{(1)}_{\text{PA}_i}$ is the output after the initial attention processing. The output of $\mathbf{F}^{(1)}_{\text{PA}_i}$ is further processed by a second self-attention mechanism

$$\mathbf{F}^{(2)}_{\text{PA}_i} = \text{Self-Attention}\left(\mathbf{F}^{(1)}_{\text{PA}_i}\right) \tag{10}$$

resulting in the refined feature map $\mathbf{F}^{(2)}_{\text{PA}_i}$ for each patch. This two-stage self-attention mechanism allows for more comprehensive feature interaction within each patch, ensuring that

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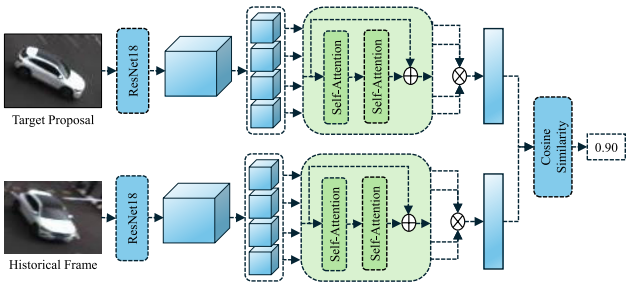


Fig. 3. Schematic of the appearance module in UBSTrack. Features are extracted from the target proposal and historical appearance using ResNet18, split into four patches, and processed through an attention mechanism comprised of two self-attention layers to capture local and global features. Cosine similarity is then used to measure the similarity between the target's appearance and the proposals.

both local and global relationships are captured effectively. After processing through the two attention layers, the final feature maps are produced using the following equation:

$$\hat{\mathbf{F}}_{\mathrm{PA}_i} = \mathbf{F}_{\mathrm{PA}_i} + \mathbf{F}^{(2)}_{\mathrm{PA}_i}. \tag{11}$$

All refined patch feature maps $\hat{\mathbf{F}}_{\mathrm{PA}_i}$ are then concatenated to form a unified feature representation

$$\mathbf{F}_{\mathrm{c}} = \left\{\hat{\mathbf{F}}_{\mathrm{PA}_1}; \hat{\mathbf{F}}_{\mathrm{PA}_2}; \hat{\mathbf{F}}_{\mathrm{PA}_3}; \hat{\mathbf{F}}_{\mathrm{PA}_4}\right\} \tag{12}$$

where [.; .] denotes the concatenation operation. After that, the cosine similarity is employed to compute the similarity between the attention features of the proposal and each historical frame

$$\mathrm{Sim}\big((\mathbf{F}_{\mathrm{c}})_{\mathrm{pi}}, (\mathbf{F}_{\mathrm{c}})_{\mathrm{hj}}\big) = \frac{(\mathbf{F}_{\mathrm{c}})_{\mathrm{pi}} \cdot (\mathbf{F}_{\mathrm{c}})_{\mathrm{hj}}}{\|(\mathbf{F}_{\mathrm{c}})_{\mathrm{pi}}\| \|(\mathbf{F}_{\mathrm{c}})_{\mathrm{hj}}\|}. \tag{13}$$

Here, $\|\cdot\|$ represents the Euclidean norm. This similarity score measures the degree of alignment between the proposal and the target's historical appearance. For each candidate proposal $\mathbf{p}_i$, the similarity scores across all historical frames are averaged to obtain a single score

$$\mathrm{AvgSim}(\mathbf{p}_i) = \frac{1}{M}\sum_{j=1}^{M} \mathrm{Sim}\big((\mathbf{F}_{\mathrm{c}})_{\mathrm{pi}}, (\mathbf{F}_{\mathrm{c}})_{\mathrm{hj}}\big). \tag{14}$$

Finally, the candidate proposal with the highest average similarity score is selected as the most reliable proposal

$$\mathcal{P}_{\mathrm{best}} = \arg\max_{\mathbf{p}_i} \mathrm{AvgSim}(\mathbf{p}_i). \tag{15}$$

The appearance module, with its combination of feature extraction, self-attention, and similarity measurement, ensures that the selected proposal maintains temporal consistency with the target's historical appearance, thereby enhancing the robustness of the tracking system.

Historical frame generation: The historical frame generation stage is responsible for continuously updating the target's appearance information over time, thereby ensuring temporal coherence and enhancing tracking robustness. Upon selecting the optimal proposal $\mathcal{P}_{\mathrm{best}}$, the GATE mechanism is employed to regulate the integration of features from $\mathcal{P}_{\mathrm{best}}$ into the historical frame buffer queue. This update process allows the model to retain an accurate representation of the target's appearance, enabling it to handle variations in object appearance and environmental conditions effectively.

However, when the feature space is large, directly updating the historical frame buffer can significantly increase computational costs and model complexity, posing a challenge for real-time tracking. To address this issue, the GATE mechanism employs a heuristic strategy to efficiently update the historical frame buffer while maintaining the integrity of the tracking model. The intersection over union (IoU) between $\mathcal{P}_{\mathrm{best}}$ and proposals from different models is calculated to determine the reliability of the historical frame inclusion. This process is represented mathematically as follows:

$$\mathrm{IoU}_{\mathrm{sum}} = \mathrm{IoU}_{\mathrm{T}} + \mathrm{IoU}_{\mathrm{D}} \tag{16}$$

$$\mathrm{IoU}_{\mathrm{T}} = \mathrm{IoU}(\mathcal{P}_{\mathrm{T}}, \mathcal{P}_{\mathrm{best}}) \tag{17}$$

$$\mathrm{IoU}_{\mathrm{D}} = \max_{\mathrm{j}}\big(\mathcal{P}_{\mathrm{D}_j}, \mathcal{P}_{\mathrm{best}}\big) \tag{18}$$

where $\mathcal{P}_{\mathrm{T}}$ and $\mathcal{P}_{\mathrm{D}}$ represent the proposals generated by tracker and detector.

After that, the GATE mechanism updates the historical frame buffer if the combined IoU exceeds a predefined threshold τ

$$\text{If } \mathrm{IoU}_{\mathrm{sum}} > \tau, \text{ then update: } \mathcal{H} \leftarrow \mathrm{Add}(\mathbf{F}_{\mathrm{best}}) \tag{19}$$

where, $\mathcal{H}$ denotes the historical frame buffer queue and $\mathbf{F}_{\mathrm{best}}$ represents the feature map associated with $\mathcal{P}_{\mathrm{best}}$.

D. Training Loss

In our experiments, two distinct loss functions are utilized to optimize the proposed framework: reconstruction loss ($\mathcal{L}_r$) for UBS and triplet loss ($\mathcal{L}_{\mathrm{Triplet}}$) for the appearance module. The loss functions for the detector and tracker are adopted from [51] and [61], respectively.

Reconstruction loss: The reconstruction loss is formulated to quantify the difference between the input HVF ($\mathbf{X}$) and the reconstructed HVF ($\hat{\mathbf{X}}$), ensuring that the model effectively preserves critical spectral and spatial information during the band selection and reconstruction process. The loss function combines the mean squared error ($\mathcal{L}_{\mathrm{mse}}$) and L_1 loss ($\mathcal{L}_{L_1}$) in a weighted manner, as defined below

$$\mathcal{L}_r = \alpha \cdot \mathcal{L}_{\mathrm{mse}} + \beta \cdot \mathcal{L}_{L_1} \tag{20}$$

$$\mathcal{L}_{\mathrm{mse}} = \frac{1}{n}\sum_{i=1}^{n}\big(\mathbf{X}_i - \hat{\mathbf{X}}_i\big)^2 \tag{21}$$

$$\mathcal{L}_{L_1} = \frac{1}{n}\sum_{i=1}^{n}\big|\mathbf{X}_i - \hat{\mathbf{X}}_i\big| \tag{22}$$

where n is the number of pixels in the HVF, and α and β are weights used to balance the contributions of the two-loss components. In our experiments, both α and β are set to 0.5 to ensure equal importance is given to the pixel-wise and absolute differences during optimization.

Triplet Loss: The triplet loss function used in the appearance module is designed to enhance the discriminative capability of the model by ensuring that the embeddings of similar objects (anchor-positive pairs) are closer in feature space, while those of dissimilar objects (anchor-negative pairs) are farther apart. If $\mathbf{a}$, $\mathbf{p}$, and $\mathbf{q}$ are the feature representations of the target object in the current frame (anchor), the target object in the

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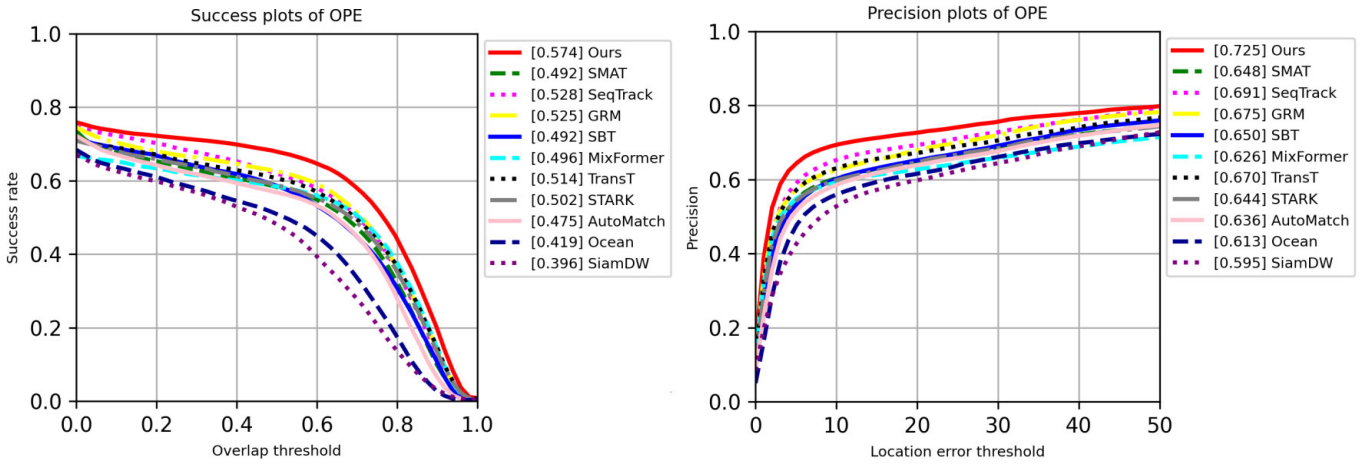


Fig. 4. Comparison of RGB-based trackers in terms of success and precision plots.

TABLE I
NUMBER OF TRAINING AND VALIDATION VIDEOS

Video Category	# of Bands	Training		Validation	
		# videos	# frames	# videos	# frames
nir	25	70	13017	30	5049
rednir	15	36	15386	20	8872
vis	16	111	44776	67	37809
Total		217	73179	117	51730

historical frame (positive), and a distractor or background object (negative), respectively, the Triplet Loss function is defined as

$$\mathcal{L}_{\text{Triplet}} = \max(0, \text{Sim}(\mathbf{a}, \mathbf{q}) - \text{Sim}(\mathbf{a}, \mathbf{p}) + l) \tag{23}$$

where l is the margin; in our experiment, the value is 1.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents an in-depth discussion of the experimental setup, the results of ablation studies, and a comparative performance analysis against state-of-the-art hyperspectral object trackers.

A. Experimental Setup

1) Datasets: In this study, we conducted experiments using the HOT2024 dataset. This dataset was introduced as part of the HOT challenges.[1] The dataset was acquired using three XIMEA snapshot cameras integrated with an IMEC hyperspectral sensor, capturing data at 25 frames per second (FPS). The hyperspectral video frames were initially recorded in a 2-D mosaic format. Additionally, the datasets include HSV-FalseColor videos, which provide a converted visualization of the original hyperspectral data into the RGB domain, enabling easier interpretation of the scene. The number of training and validation videos in the datasets is shown in Table I.

2) Implementation Details: All experiments for the UBSTrack framework were conducted on a workstation equipped with an Intel Core i7-12700 processor, 32 GB of RAM, and an NVIDIA RTX A4000 GPU with 16 GB of dedicated graphics memory. The base model, OSTrack [51], was adopted directly from its original implementation, serving as a robust foundation for object-tracking tasks. Additionally, YOLO [61] was trained following the methodology outlined in [11], ensuring seamless integration within the UBSTrack framework.

The UBS and appearance modules were trained using the Adam optimizer with a learning rate of 0.001. The models were trained for 100 epochs to ensure convergence and optimal performance across a variety of hyperspectral tracking scenarios.

The performance of the proposed method was evaluated using multiple metrics, including the area under the curve (AUC), DP@20 pixels, precision, and success plots. To assess the robustness of UBSTrack, a comparative analysis was performed against 17 state-of-the-art methods, comprising ten RGB-based trackers and seven HOT trackers. Notably, the RGB-based trackers were evaluated on HSI-FalseColor videos using the same machine, ensuring consistency and fairness in the evaluation process.

B. Comparison With RGB-Based Trackers

The comparison of UBSTrack with 10 state-of-the-art RGB-based trackers is presented in Table II, while the corresponding precision and success plots are illustrated in Fig. 4. The results reveal that UBSTrack achieves the highest AUC score of 0.574, outperforming all competing methods. This significant performance demonstrates UBSTrack's superior ability to maintain accurate overlap with ground-truth bounding boxes across various overlap thresholds. Among the RGB-based trackers, SeqTrack [70] (AUC: 0.528) and GRM [71] (AUC: 0.525) rank second and third, respectively. Despite their relatively high AUC scores, both methods exhibit a noticeable gap compared to UBSTrack, underscoring reduced robustness in handling complex tracking scenarios.

In terms of precision, UBSTrack again leads with a top precision rate of 0.725, indicating its superior accuracy in maintaining consistent object localization while minimizing center location error. SeqTrack [70] (precision: 0.691) and GRM [71] (precision: 0.675) follow as the next best performers, though both lag behind UBSTrack by a significant margin. SBT (precision: 0.650) and SMAT [72] (precision: 0.648)

[1]https://www.hsitracking.com/

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TABLE II

PERFORMANCE COMPARISON OF UBSTRACK AGAINST VARIOUS STATE-OF-THE-ART RGB AND HSI TRACKERS EVALUATED ON THE HOT2024 DATASETS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

#	Tracker	Venue	Year	Backbone/Methods	AUC	DP@20Pixels	FPS	Device
1	SiamDW [63]	CVPR	2019	CIResNet-22	0.396	0.595	110.0	GPU, RTX A4000
2	Ocean [64]	ECCV	2020	ResNet50	0.419	0.613	91.3	GPU, RTX A4000
3	STARK [65]	ICCV	2021	vanilla ResNet	0.502	0.644	82.4	GPU, RTX A4000
4	AutoMatch [66]	ICCV	2021	ResNet50	0.514	0.636	81.6	GPU, RTX A4000
5	TransT [67]	CVPR	2021	ViT-Base	0.514	0.670	82.7	GPU, RTX A4000
6	MixFormer [68]	CVPR	2022	MixFormer-ViT	0.496	0.626	56.7	GPU, RTX A4000
7	SBT [69]	CVPR	2022	SBT-Base	0.492	0.650	34.8	GPU, RTX A4000
8	SeqTrack [70]	CVPR	2023	ViT-Large	0.528	0.691	31.3	GPU, RTX A4000
9	GRM [71]	CVPR	2023	ViT-Base	0.525	0.675	28.1	GPU, RTX A4000
10	SMAT [72]	WACV	2024	MobileViTv2	0.492	0.648	120.0	GPU, RTX A4000
11	BABS [16]	GRSL	2023	VITAL	0.381	0.576	4.2	GPU, RTX A4000
12	SEE-Net [73]	TIP	2023	ResNet50	0.423	0.607	8.7	GPU, TITAN Xp
13	SENSE [57]	IF	2024	ResNet50	0.377	0.504	15.4	GPU, RTX 4060
14	PHTrack [28]	TGRS	2024	ResNet50	0.353	0.485	15.2	GPU, 2 X RTX 3090
15	SPIRIT [56]	TGRS	2024	ResNet50	0.415	0.534	26.0	GPU, RTX4060
16	MMF-Net [74]	TGRS	2024	GoogleNet	0.522	0.683	6.1	GPU, Tesla V100
17	Trans-DAT [58]	TCSVT	2024	ResNet50	0.450	0.587	124.8	GPU, RTX 4090
18	**UBSTrack**	-	-	Custom	0.574	0.725	15.7	GPU, RTX A4000

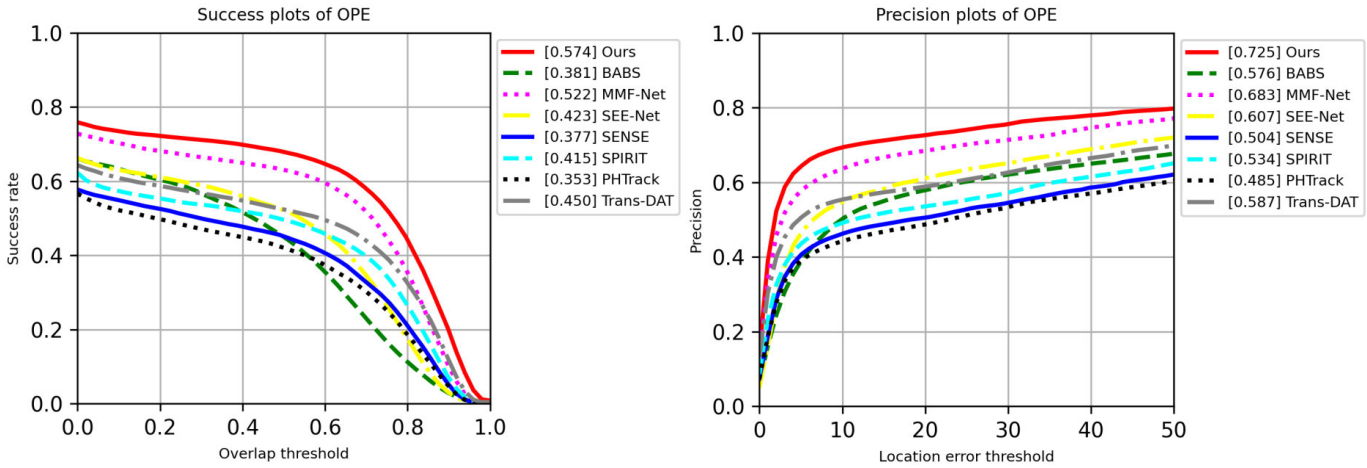


Fig. 5. Comparison of HOT trackers in terms of success and precision plots.

show moderate performance, but their precision values remain below 0.7, reflecting higher location errors. At the lower end of the spectrum, SiamDW (precision: 0.595) delivers the least accurate performance, highlighting its limitations in reliably localizing objects under challenging conditions.

The superior performance of UBSTrack over RGB-based trackers can be attributed to its ability to leverage fine-grained spectral information across a broad range of wavelengths, unlike RGB trackers, which are constrained to only three color channels. By employing a UBS strategy in combination with an appearance module with historical frames, UBSTrack effectively captures the spectral and temporal dynamics of the target, resulting in robust and consistent tracking performance across varying scenarios.

C. Comparison With HOT Trackers

The comparison of UBSTrack on seven recent state-of-the-art HOT trackers is presented in Table II and the precision and success plots on Fig. 5. We observe that UBSTrack maintains a consistently higher success rate across all overlap thresholds and achieves the highest AUC of 0.574, significantly outperforming all other methods, highlighting its ability to precisely track objects with varying appearances and motion patterns. The second-best method, MMF-Net, achieves an AUC of 0.522, followed by Trans-DAT with an AUC of 0.450. Despite being the closest competitors, both methods exhibit a noticeable performance gap compared to UBSTrack, indicating their reduced robustness in handling complex scenarios such as object occlusion or rapid motion. SEE-Net [73] (AUC: 0.423) and SPIRIT [56] (AUC: 0.415) show moderate performance but fall significantly behind UBSTrack, while BABS [16] and PHTrack [28] rank among the lower-performing methods, reflecting their limited ability to maintain accurate overlap in challenging environments.

Similarly, in precision, UBSTrack achieves the highest precision rate of 0.725, demonstrating its superior capability to accurately localize objects with minimal location error across frames. MMF-Net [74] ranks second with a precision of 0.683, followed by SEE-Net [73] (0.607) and Trans-DAT [58] (0.587). Despite their competitive performance,

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TABLE III

AUC RESULTS OF 14 ATTRIBUTES ON THE HOT-2024 DATASET. THE ATTRIBUTES INCLUDE BACKGROUND CLUTTER (BC), CAMERA MOTION (CM), DEFORMATION (DEF), FAST MOTION (FM), IN-PLANE ROTATION (IPR), ILLUMINATION VARIATION (IV), LOW LIGHT (LL), LOW RESOLUTION (LR), MOTION BLUR (MB), OCCLUSION (OCC), OUT-OF-PLANE ROTATION (OPR), OUT-OF-VIEW (OV), REFLECTIONS (REF), AND SCALE VARIATIONS (SV). THE TOP TWO VALUES ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

	SiamDW [63]	Ocean [64]	AutoMatch [66]	TransT [67]	SBT [69]	GRM [71]	SMAT [72]	BABS [16]	SEE-Net [73]	SENSE [57]	PHTrack [28]	SPIRIT [56]	MMF-Net [74]	Trans-DAT [58]	Ours
BC	0.291	0.328	0.326	0.397	0.391	0.402	0.354	0.338	0.333	0.317	0.295	0.346	0.393	0.363	0.475
CM	0.195	0.163	0.229	0.392	0.402	0.470	0.181	0.192	0.274	0.303	0.286	0.366	0.247	0.388	0.352
DEF	0.296	0.342	0.293	0.354	0.397	0.273	0.303	0.313	0.227	0.211	0.198	0.246	0.444	0.469	0.481
FM	0.468	0.505	0.629	0.588	0.576	0.646	0.646	0.521	0.511	0.571	0.545	0.641	0.653	0.538	0.725
IPR	0.424	0.505	0.516	0.592	0.556	0.619	0.544	0.423	0.438	0.405	0.379	0.389	0.573	0.443	0.628
IV	0.271	0.384	0.412	0.482	0.453	0.356	0.465	0.324	0.378	0.306	0.101	0.401	0.402	0.464	0.517
LL	0.402	0.508	0.580	0.707	0.638	0.654	0.672	0.343	0.529	0.409	0.267	0.537	0.647	0.526	0.785
LR	0.188	0.099	0.120	0.277	0.240	0.219	0.180	0.188	0.221	0.160	0.191	0.210	0.170	0.239	0.343
MB	0.044	0.031	0.035	0.173	0.150	0.047	0.026	0.003	0.035	0.002	0.002	0.001	0.068	0.084	0.186
OCC	0.374	0.409	0.484	0.496	0.480	0.519	0.478	0.372	0.405	0.374	0.349	0.405	0.516	0.441	0.596
OPR	0.322	0.504	0.545	0.575	0.579	0.591	0.530	0.359	0.480	0.393	0.288	0.474	0.553	0.526	0.638
OV	0.271	0.269	0.242	0.248	0.234	0.381	0.257	0.307	0.312	0.299	0.337	0.306	0.348	0.275	0.372
REF	0.067	0.050	0.046	0.229	0.070	0.166	0.039	0.045	0.139	0.078	0.136	0.089	0.175	0.292	0.223
SV	0.491	0.539	0.658	0.614	0.597	0.664	0.655	0.494	0.515	0.526	0.480	0.596	0.697	0.572	0.725

TABLE IV

PRECISION (DP@20PIXELS) RESULTS OF 14 ATTRIBUTES ON THE HOT 2024 DATASET. THE TOP TWO VALUES ARE HIGHLIGHTED IN RED AND BLUE, RESPECTIVELY

	SiamDW [63]	Ocean [64]	AutoMatch [66]	TransT [67]	SBT [69]	GRM [71]	SMAT [72]	BABS [16]	SEE-Net [73]	SENSE [57]	PHTrack [28]	SPIRIT [56]	MMF-Net [74]	Trans-DAT [58]	Ours
BC	0.457	0.438	0.415	0.511	0.506	0.539	0.477	0.523	0.465	0.408	0.409	0.432	0.523	0.473	0.601
CM	0.293	0.239	0.322	0.579	0.575	0.660	0.254	0.296	0.401	0.416	0.392	0.512	0.370	0.554	0.477
DEF	0.430	0.384	0.272	0.422	0.456	0.272	0.281	0.373	0.205	0.215	0.168	0.243	0.502	0.583	0.641
FM	0.739	0.784	0.832	0.733	0.746	0.823	0.840	0.834	0.795	0.759	0.744	0.799	0.826	0.654	0.867
IPR	0.587	0.659	0.638	0.712	0.666	0.761	0.683	0.633	0.557	0.496	0.496	0.463	0.716	0.536	0.760
IV	0.319	0.419	0.498	0.576	0.610	0.418	0.560	0.211	0.428	0.386	0.082	0.458	0.483	0.629	0.633
LL	0.493	0.580	0.686	0.789	0.733	0.714	0.732	0.297	0.640	0.539	0.324	0.601	0.747	0.601	0.904
LR	0.325	0.212	0.233	0.434	0.387	0.397	0.349	0.344	0.406	0.248	0.290	0.329	0.307	0.369	0.506
MB	0.126	0.338	0.372	0.422	0.327	0.315	0.357	0.040	0.121	0.007	0.007	0.016	0.198	0.189	0.429
OCC	0.569	0.596	0.634	0.633	0.622	0.666	0.616	0.560	0.580	0.495	0.478	0.512	0.662	0.564	0.744
OPR	0.441	0.630	0.676	0.690	0.676	0.709	0.642	0.462	0.589	0.480	0.361	0.545	0.683	0.637	0.760
OV	0.429	0.388	0.312	0.298	0.295	0.526	0.327	0.566	0.452	0.411	0.467	0.467	0.474	0.351	0.480
REF	0.179	0.137	0.131	0.405	0.186	0.337	0.099	0.203	0.275	0.173	0.251	0.245	0.333	0.477	0.368
SV	0.747	0.791	0.853	0.752	0.768	0.817	0.817	0.738	0.749	0.704	0.657	0.741	0.861	0.718	0.867

these methods still exhibit a considerable gap when compared to UBSTrack, indicating less reliable object localization. SENSE [57] (0.504), SPIRIT [56] (0.534), and PHTrack [28] (0.485) show relatively lower precision rates, highlighting their limited accuracy in tracking under challenging conditions.

The success of UBSTrack can be attributed to its innovative UBS mechanism and the seamless integration of a historical appearance module. Unlike conventional HOT trackers that either rely on fixed bands or hand-crafted features, UBSTrack dynamically selects the most informative spectral bands using a data-driven approach, ensuring adaptability across varying object appearances and environmental conditions. Additionally, by incorporating a historical appearance module, UBSTrack enhances temporal coherence and mitigates issues arising from object occlusion, rapid motion, and significant appearance changes. This strategic combination of spectral and temporal features enables UBSTrack to deliver superior tracking performance, making it highly robust and reliable for real-world HOT scenarios.

D. Attribute Analysis

We compare UBSTrack against state-of-the-art methods across 14 challenging attributes on the HOT2024 dataset. The results, presented in Tables III and IV, demonstrate that UBSTrack consistently achieves superior performance in terms of both precision and success. Notably, it achieves the highest AUC scores in key attributes such as FM with 0.725, OPR with 0.638, and IV with 0.517, highlighting its robustness in handling complex backgrounds, dynamic object movements, and illumination variation. Similarly, in precision, UBSTrack consistently achieves the highest precision across the same attributes, such as FM, with 0.867, OPR with 0.760, and IV with 0.633, further validating its ability to handle intricate tracking scenarios effectively.

UBSTrack also excels in addressing particularly challenging attributes like OCC and BC, achieving AUC scores of 0.596 and 0.475, and precision scores of 0.744 and 0.601, respectively. While competing methods such as GRM [71] and SiamDW [63] perform well in specific attributes such as CM

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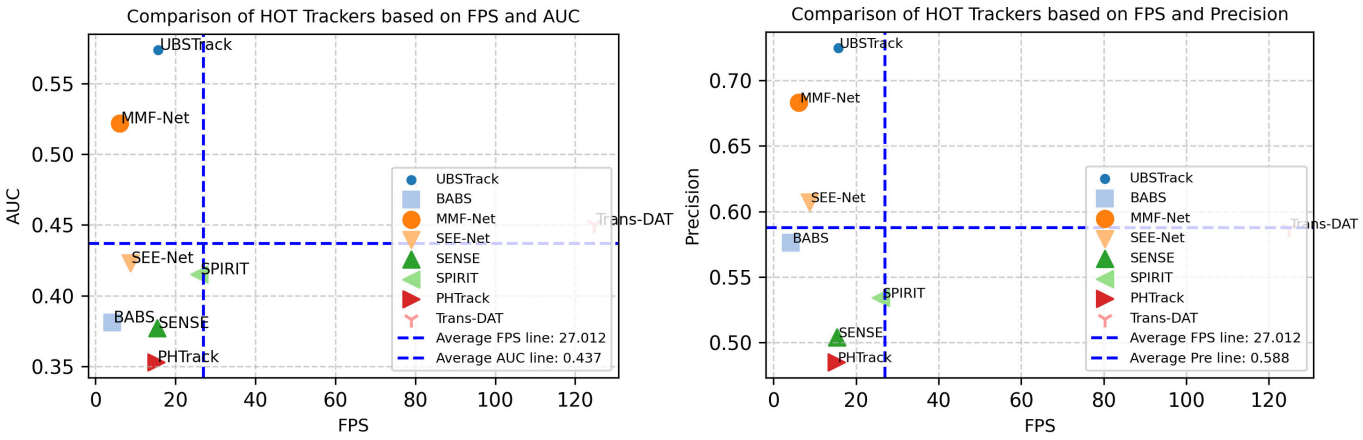


Fig. 6. Efficiency comparison with HOT trackers.

and OV, UBSTrack consistently outperforms them across most attributes, underscoring its versatility and adaptability. These results firmly establish UBSTrack as a robust and reliable solution for HOT in diverse real-world conditions.

E. Accuracy Versus Efficiency

Fig. 6 illustrates the trade-off between accuracy and efficiency for various HOT trackers, using AUC (left) and precision (right) metrics plotted against FPS. The vertical dashed blue line represents the average FPS of 27.012, while the horizontal dashed blue lines mark the average AUC (0.437) and precision (0.588), respectively. This comparison highlights the relative performance of each tracker in terms of balancing tracking accuracy and computational speed.

In the AUC plot (left), UBSTrack achieves the highest AUC of 0.574 with an FPS of approximately 15.7. While slightly below the average FPS, UBSTrack's accuracy is well above all other methods, indicating its ability to deliver superior tracking performance with moderate computational cost. MMF-Net follows with the second-highest AUC of 0.522 but at a much lower FPS (6.1), which places it far from the average FPS line. Most other trackers, including SEE-Net [73], SPIRIT [56], BABS [16], and PHTrack [28], exhibit both lower AUC scores and FPS values, indicating inferior performance across both metrics. Notably, Trans-DAT [58] achieves the highest FPS of over 120, far exceeding all other methods, but with a low AUC of 0.450, favoring speed at the expense of accuracy.

A similar trend is observed in the precision plot (right), where UBSTrack once again leads with the highest precision of 0.725, confirming its reliability in accurate object localization. Operating at an FPS of 15.7, UBSTrack demonstrates a well-balanced trade-off between precision and speed. MMF-Net [74] ranks second in precision (0.683) but operates at a significantly lower FPS, indicating a less efficient performance compared to UBSTrack. SEE-Net [73] and BABS [16] achieve moderate precision values (0.607 and 0.576, respectively) but remain below average in terms of FPS, reflecting reduced efficiency. Conversely, Trans-DAT [58] stands out with the highest FPS, exceeding 120, but delivers a low precision of 0.587, reiterating its prioritization of speed over accuracy.

These results clearly establish UBSTrack as the most balanced tracker among competing HOT methods, offering both high accuracy and near-average computational efficiency. By maintaining the highest AUC and precision scores while operating at an acceptable FPS, UBSTrack proves to be an optimal solution for real-time HOT, where both precision and efficiency are crucial.

F. Qualitative Analysis

Fig. 7 highlights the performance of UBSTrack compared to state-of-the-art HOT trackers on challenging scenarios in the HOT2024 dataset. In sequence (nir-rider10) [Fig. 7(a)], where attributes such as OCC and SV are present, UBSTrack effectively maintains precise tracking despite partial occlusions and scale changes, outperforming competing trackers that either drift or lose the target entirely. In sequence (render-dice3) [Fig. 7(b)], while other trackers mistakenly follow a similar distractor dice after occlusion, the proposed UBSTrack framework successfully re-identifies and follows the correct target, demonstrating superior robustness and adaptability to complex scenarios. Similarly, in sequence (vis-backpack4) [Fig. 7(c)] with deformation and objects with similar color, UBSTrack distinguishes the target effectively, avoiding drift, and in sequence (vis-cards6) [Fig. 7(d)], UBSTrack consistently tracks the target even under complex scenarios. These results underscore UBSTrack's adaptability and superior performance in diverse and challenging tracking scenarios.

G. Scalability and Generalizability of UBSTrack for Real-World UAV Dataset

In this section, we discuss the ability of UBSTrack to handle the diverse challenges associated with real-world UAV-acquired data. Currently, there is no publicly available hyperspectral dataset specifically captured by UAV platforms. Therefore, we employed the RGB-based UAV dataset [75], which provides high-resolution RGB video sequences captured by UAVs. This dataset is particularly suitable for validating the performance of UBSTrack in dynamic UAV scenarios, where real-world challenges such as background clutter, object occlusions, and varying motion dynamics are prevalent.

The RGB-based UAV dataset, however, lacks hyperspectral information, making it incompatible with most HOT frameworks that rely on spectral data for feature extraction and object tracking. In contrast, UBSTrack's UBS technique is inherently flexible, enabling it to operate on data of varying modalities, including RGB datasets. This adaptability allows

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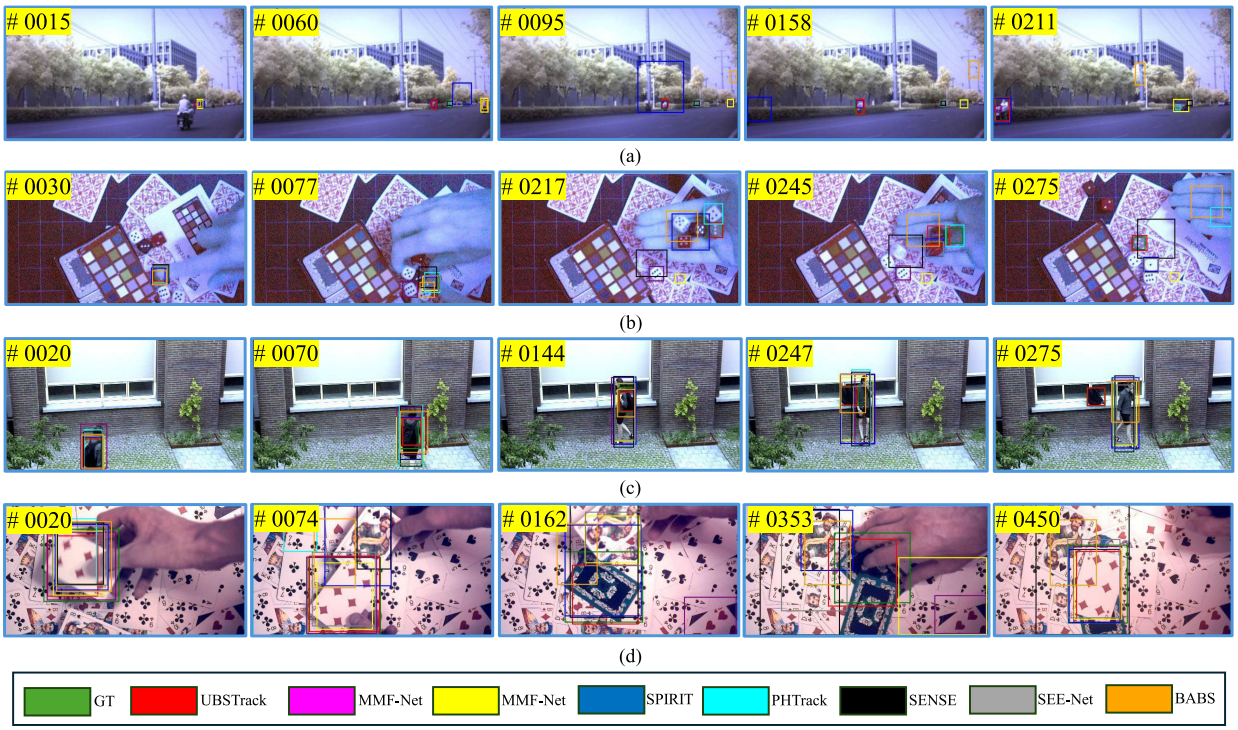


Fig. 7. Qualitative analysis of UBSTrack with the HOT trackers. (a) nir-rider10, attribute: OCC, SV. (b) rednir-dice3, attribute: OCC, LC, BC, IPR, OPR. (c) vis-backpack4, attribute: DEF, OCC, OSC. (d) vis-cards6: BC, OPR, OCC, IPR, OSC.

UBSTrack to demonstrate its generalizability and robustness even in the absence of hyperspectral data.

Fig. 8 illustrates the performance of UBSTrack across four scenarios from the UAV dataset. In scenario S1701 [Fig. 8(a)], where the scene involves background clutter and variations in object motion, UBSTrack demonstrates its robustness by consistently identifying and tracking the target objects across all frames. This showcases the framework's ability to operate effectively in cluttered environments with dynamic object interactions. In S1601 [Fig. 8(b)], the scene introduces significant motion dynamics, including high-speed object movements and variations in perspective caused by the UAV's flight path. Despite these challenges, UBSTrack maintains temporal consistency and accurately tracks the vehicle throughout the sequence, highlighting its adaptability to challenging motion conditions.

In S0310 [Fig. 8(c)], the tracking task is further complicated by occlusions caused by vegetation and variations in the UAV's altitude. UBSTrack handles these challenges effectively by re-identifying the target after partial occlusions, ensuring reliable tracking across most frames. However, slight inconsistencies are observed in frames #0067 and #0085, where the appearance of the tracked object deviates slightly from previous frames. These deviations may be attributed to variations in object appearance caused by occlusion artifacts or changes in perspective. While UBSTrack largely maintains consistency throughout the sequence, these observations highlight potential areas for improvement in handling subtle appearance variations during tracking.

H. Ablation Study

This section presents the ablation study to evaluate the contributions of key components in the proposed UBSTrack framework, focusing on the impact of the UBS module and the integration of the ensemble module.

TABLE V
ABLATION STUDY: SIGNIFICANT ON TOP-THREE BANDS

# of Frame buffer	AUC	DP@20pixels
1	0.574	0.725
2	0.549	0.701
3	0.559	0.708
4	0.557	0.703
5	0.533	0.678

1) Band Significance Analysis: In this study, we analyze the significance of the first pseudocolor image generated by our framework, which is composed of the three most informative spectral bands selected through the proposed UBS mechanism. To evaluate its effectiveness, we conducted an ablation study using the top-5 pseudocolor images, ranked by the UBS module based on their attention scores. The results, presented in Table V, show that the first pseudocolor image yields the highest tracking performance, achieving an AUC of 0.574 and a precision score of 0.725. In comparison, the second to fifth-ranked pseudocolor combinations result in slightly lower performance, with AUCs ranging from 0.533 to 0.559 and precision scores between 0.678 and 0.708. These results validate the effectiveness of the UBS module in selecting the top three spectral bands that possess strong spatial–spectral discriminability, thereby enhancing the robustness and accuracy of the tracking framework.

2) Invariance of UBS to Background Variations: In this ablation study, we evaluate whether the spectral bands selected by the proposed UBS module remain informative and effective across varying background conditions within a video sequence. Specifically, we aim to assess whether the fixed

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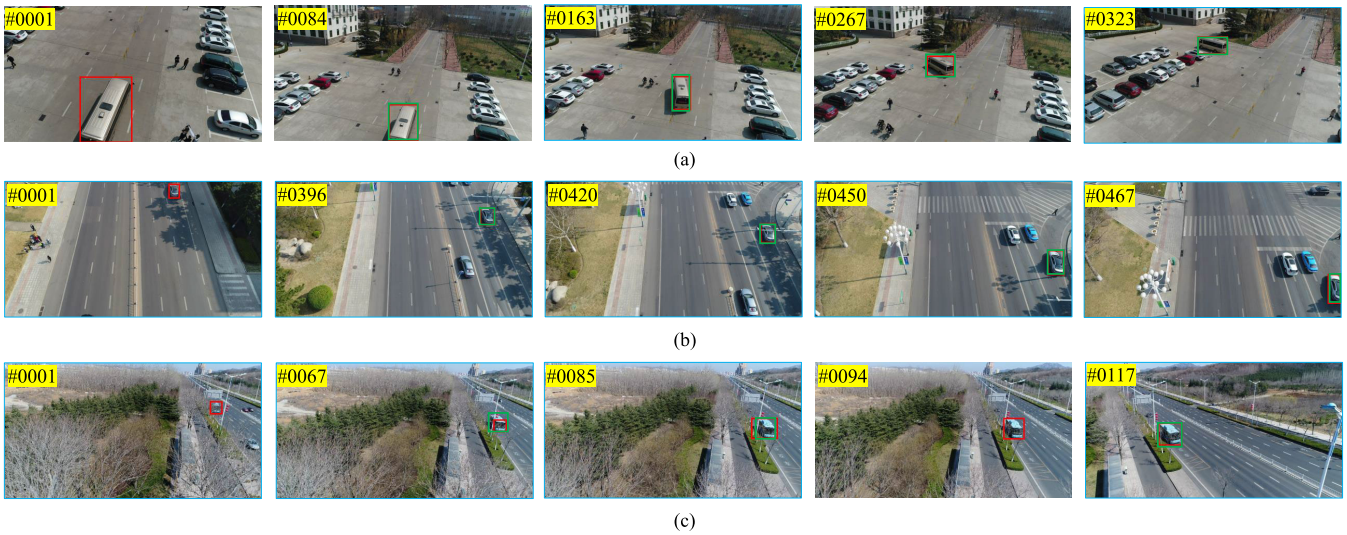


Fig. 8. Demonstration of UAV-based object tracking scenarios. The figure illustrates typical challenges encountered in UAV data, such as varying altitudes, dynamic backgrounds, object occlusion, and high-speed movements, highlighting the applicability of UBSTrack in real-world UAV environments. We consider three different videos from the UAV dataset [75]. (a) S1701, (b) S1601, and (c) S0310. The green bounding box represents the ground-truth, and the red bounding box represents the UBSTrack output.

TABLE VI
COMPARISON OF VIDEO-WISE (FIXED) AND FRAME-WISE (DYNAMIC) BAND SELECTION STRATEGIES

Band Selection Strategy	AUC	DP@20Pixels	FPS
Video-wise (fixed)	0.574	0.725	15.7
Frame-wise (dynamic)	0.568	0.722	8.9

top-3 bands selected at the beginning of a sequence are robust to background changes that occur throughout the hyperspectral video frames. We employed two-band selection strategies: 1) video-wise (fixed) selection, where the same three bands are used consistently for the entire sequence and 2) frame-wise (dynamic) selection, where the UBS module selects a new set of bands independently for each frame. As shown in Table VI, the fixed-band strategy achieves superior performance, with an AUC of 0.574 and DP@20Pixels of 0.725, compared to the dynamic selection approach, which yields an AUC of 0.568 and DP@20Pixels of 0.722. Furthermore, the frame-wise strategy introduces a significant drop in computational efficiency, as reflected in reduced FPS from 15.7 to 8.9. These results indicate that the UBS module selects bands based on object-centric spectral features rather than background information. The local and global attention-based mechanisms within UBS contribute to this robustness, making the selected bands largely invariant to background changes while maintaining both accuracy and efficiency.

3) Comparison With Other Band Selection Techniques: We compare the UBS module in UBSTrack against three existing band selection techniques, as summarized in Table VII. The UBS module achieves the highest AUC score of 0.574 and DP@20Pixels of 0.725, surpassing all competing methods. SiamOHOT [27] relies on fixed bands for pseudocolor image generation, limiting its adaptability. BABS [16] incorporates spatial–spectral differences using hand-crafted features, which fail to fully capture object-specific characteristics. BAE-Net [17], while employing a band attention module, requires separate models for datasets with varying numbers of bands, reducing its generalization capability. In contrast, the UBS module unifies band selection into a single model, leveraging both spatial and spectral information effectively, thereby significantly enhancing tracking performance.

TABLE VII
ABLATION STUDY: COMPARISON WITH OTHER BAND SELECTION TECHNIQUES

	AUC	ΔAUC	DP@20Pixels	ΔDP@20Pixels
SiamOHOT [27]	0.541	3.3%	0.684	4.1%
BABS [16]	0.560	1.4%	0.713	1.2%
BAE-Net [17]	0.558	1.6%	0.705	2.0%
UBSTrack	0.574	-	0.725	-

TABLE VIII
ABLATION STUDY: IMPACT OF APPEARANCE MODULE

	AUC	ΔAUC	DP@20Pixels	ΔDP@20Pixels
YOLO [61]	0.484	9.0%	0.620	10.5%
OSTrack [51]	0.520	5.4%	0.661	6.4%
UBSTrack	0.574	-	0.725	-

4) Impact of Appearance Module: Table VIII highlights the impact of the appearance module on the overall performance of UBSTrack by comparing it against individual components such as YOLO [61] and OSTrack [51]. UBSTrack achieves the highest AUC score of 0.574 and DP@20Pixels of 0.725, significantly outperforming individual components YOLO [61] and OSTrack [51], which achieve AUC scores of 0.484 and 0.520, respectively. The superior performance of UBSTrack is attributed to the integration of the appearance module, which combines local and global attention mechanisms to capture temporal consistency and spatial coherence. YOLO, as a standalone detector, struggles with precision due to its reliance on static object detection, resulting in lower DP@20Pixels (0.620). OSTrack [51], while more robust as a tracker, lacks the ability to integrate complementary features from detectors, resulting in an AUC of 0.520 and DP@20Pixels of 0.661.

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TABLE IX
ABLATION STUDY: IMPACT OF FRAME BUFFER SIZE

# of Frame buffer	AUC	DP@20pixels
1	0.571	0.719
2	0.570	0.719
3	0.571	0.720
4	0.574	0.725
5	0.570	0.719
6	0.570	0.720

TABLE X
ABLATION STUDY: IMPACT OF GATE MECHANISM

IOU Threshold	AUC	DP@20pixels
1.1	0.568	0.719
1.2	0.569	0.719
1.3	0.569	0.718
1.4	0.574	0.725
1.5	0.568	0.716

In contrast, UBSTrack's appearance module unifies outputs from both base models using historical information, effectively combining their strengths and mitigating their limitations to achieve superior tracking accuracy and precision.

5) Impact of Frame Buffer Size: Table IX presents an ablation study analyzing the effect of the number of frames in the frame buffer on UBSTrack's performance. The results indicate that a frame buffer size of 4 achieves the highest AUC of 0.574 and DP@20pixels of 0.725. While frame buffer sizes of 1, 3, and 5 produce comparable results with AUC values of approximately 0.570–0.571, the optimal performance is observed with a buffer size of 4. This demonstrates that a balanced frame buffer enhances tracking by improving temporal consistency, while larger buffers dilute the relevance of recent frames.

6) Impact of GATE Mechanism: The ablation study in Table X evaluates the impact of varying the IoU threshold in the GATE mechanism, which determines the inclusion of candidates in the frame buffer. An IoU threshold of 1.4 yields the highest performance, with an AUC of 0.574 and DP@20pixels of 0.725. This indicates that a properly calibrated threshold effectively balances the inclusion of relevant proposals while excluding noise, thereby optimizing tracking accuracy. Deviating from this threshold, whether lower or higher, leads to reduced performance, highlighting the importance of maintaining an optimal balance for preserving temporal coherence.

V. CONCLUSION

In this study, we propose UBSTrack, a novel HOT framework designed to effectively integrate detection and tracking models using a UBS module and a model ensemble technique leveraging the local–global attention-based appearance mechanism. UBSTrack outperforms state-of-the-art RGB and hyperspectral trackers across challenging datasets, demonstrating superior robustness, adaptability, and efficiency. By leveraging the UBS module, the framework efficiently achieves optimal band selection, preserving critical spectral–spatial information while remaining independent of specific spectral characteristics of the dataset. The integration of the GATE mechanism and frame buffer further enhances temporal coherence, while the ensemble of detection and tracking models provides robust target localization under diverse environmental conditions. Future work can explore extending the framework to handle multimodal data, further optimizing computational efficiency for real-time UAV applications, and enhancing performance in challenging conditions, such as low-light environments and high-speed object motion.

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